Medical Store Management System

1. Introduction

The Medical Store Management System (MSMS) is a software application designed to manage the daily operations of a medical store. The system helps in managing medicines, stock levels, sales, purchases, suppliers, customers, and billing efficiently and accurately.

2. Objectives

- To automate the process of managing medicine inventory.
- To reduce manual errors in billing and stock management.
- To maintain accurate records of sales and purchases.
- To generate reports for business analysis.
- To improve customer service and operational efficiency.

3. Scope of the System

The system is intended for use in retail medical stores and pharmacies. It covers inventory management, supplier management, customer management, sales processing, purchase management, and report generation.

4. System Modules

4.1 Admin Module

- User management (Add, Update, Delete users)
- Role-based access control
- System configuration

4.2 Medicine Management Module

- Add new medicines
- Update medicine details
- Delete medicines
- Track expiry dates
- Monitor stock levels

4.3 Supplier Management Module

- Add supplier details
- Update supplier information
- Track supplier transactions

4.4 Customer Management Module

- Add customer details
- Maintain customer purchase history

4.5 Sales Management Module

- Generate bills
- Calculate total amount with taxes and discounts
- Print invoices
- Update stock automatically after sales

4.6 Purchase Management Module

- Record medicine purchases
- Update stock after purchase
- Manage purchase invoices

4.7 Reports Module

- Daily sales report
- Monthly sales report
- Stock report
- Expiry report

5. Functional Requirements

1. The system shall allow the admin to log in securely.
2. The system shall allow adding, updating, and deleting medicines.
3. The system shall automatically update stock after each sale and purchase.
4. The system shall generate invoices for customers.
5. The system shall generate reports for analysis.
6. The system shall alert users for low stock and near expiry medicines.

6. Non-Functional Requirements

7. The system should be user-friendly.
8. The system should provide secure authentication and authorization.
9. The system should ensure data integrity and reliability.
10. The system should respond quickly to user actions.
11. The system should support regular data backup.

7. System Requirements

7.1 Hardware Requirements

- Processor: Intel i3 or above

- RAM: Minimum 4 GB
- Hard Disk: Minimum 500 GB
- Printer for invoice printing

7.2 Software Requirements

- Operating System: Windows / Linux
- Frontend: HTML/CSS/JavaScript (if web-based)
- Backend: PHP / Python / Java
- Database: MySQL / PostgreSQL

8. Database Design

Users Table

- user_id (Primary Key)
- username
- password
- role

Medicines Table

- medicine_id (Primary Key)
- name
- category
- batch_number
- expiry_date
- price
- quantity

Suppliers Table

- supplier_id (Primary Key)
- name
- contact
- address

Sales Table

- sale_id (Primary Key)
- customer_name
- date
- total_amount

9. Advantages of the System

- Reduces paperwork and manual errors.

- Improves efficiency and productivity.
- Provides real-time stock monitoring.
- Ensures better customer service.

10. Conclusion

The Medical Store Management System is an essential tool for modern pharmacies. It enhances efficiency, accuracy, and profitability by automating key business operations. The system ensures proper inventory control, smooth billing processes, and effective record management.